

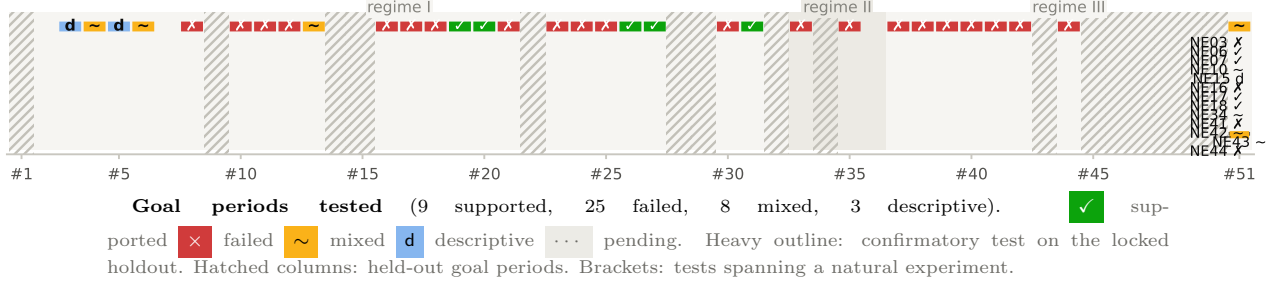
# H39: Catalysts vs. fields

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 2/5 Usefulness 2.5/5

**Bottom line.** No operator lever is a pure catalyst: nudges both unstick and tilt agents, while human messages, mentions, erasures and goals act mainly as fields, on activity or on content depending on the regime.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



## WHAT HAPPENED

Which operator levers change *what* agents do (fields: stationary occupancy shifts) and which only speed them up (catalysts: escape at fixed occupancy)? Matched-window kinetic decomposition of behavior-state chains around nudges, human messages, @-mentions and context erasures in 32 non-holdout periods, plus step levers. **Round 1b** used the nudge's leading @ as its only target, DQ8's presence-cut lever design, and Jev v3.1 behavior states as a second state space (V4). **Nudges are both**, now in two state spaces: B4  $K = +0.13$ , idle share  $-0.048$  (CI now excludes 0); V4  $K = +0.25$ , waiting share  $-0.08$ , waiting escape  $\times 1.6$ . Human messages are fields; @-mentions steer content ( $+0.33$  SD), not behavior. Forced erasure is a field toward work in both state spaces (its  $K$  flips sign between them). Natives: switching the nudger off is a catalytic loss in both state spaces (NE43); nudges are not more catalytic under the 12-h pause default (NE44). Holdout not run.

## PHYSICAL MODEL

Models 02/10: a kinetic Potts (Markov-jump) agent over states  $s \in \{\text{work, chat, idle, consolidate}\}$  with Arrhenius rates  $k_{ij}$ , well depths  $G_i$  and symmetric barriers  $B_{ij}$ . A field  $h$  tilts the wells; a catalyst  $c$  lowers the barriers. Measured per lever: kicked and control transition matrices  $T^K, T^C$  in matched 30-min windows, their stationary distributions  $\pi^K, \pi^C$ , the escape probability  $e_s = 1 - T_{ss}$ , the field index  $\phi$ , the catalytic index  $K$ , and the catalytic fraction  $\rho$ .

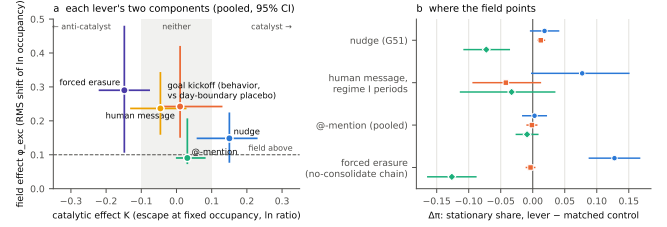
$$k_{ij} = \nu e^{-(B_{ij}-G_i)/T}, \quad \pi_i \propto e^{-G_i/T}$$

$$s_{ij} = \frac{1}{2}(\delta \ln k_{ij} + \delta \ln k_{ji}) \quad (\text{catalytic part})$$

$$a_{ij} = \frac{1}{2}(\delta \ln k_{ij} - \delta \ln k_{ji}) \quad (\text{field part})$$

$$\phi = \text{SD}_{\pi^C} \left[ \ln \pi_s^K - \ln \pi_s^C \right], \quad \rho = |K| / (|K| + \phi_{\text{exc}})$$

$$K = \ln \left( \sum_s \pi_s^C e_s^K / \sum_s \pi_s^C e_s^C \right)$$



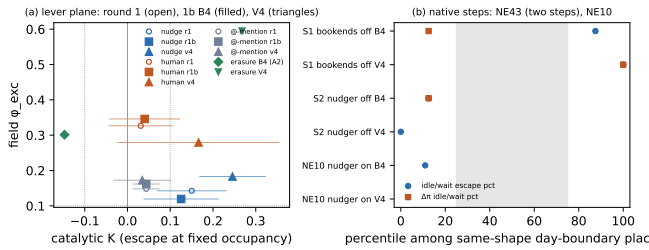
**Observables.** (a) Pooled field strength  $\phi_{\text{exc}}$  (placebo-corrected) against catalytic effect  $K$ , with 95% CIs. Dashed line and grey band: the 0.10 thresholds. Only the nudge reaches the “both” corner; human messages and erasure are fields, and erasure is anti-catalytic. (b) Where each field points: stationary share changes for work, chat and idle.

## SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Partly useful. An operator gets a concrete rule. To unstick an idle agent, send one directed nudge: escape rises  $\sim 50\%$ , and the agent also stays out of idle while it acts. To steer what an agent writes without changing how busy it is, mention it. To change topics, set a goal; that works in the large regime-III village. The physics label is softer than it sounds: the field/catalyst split depends on a convention, the nudge verdict rests on one period (G51), and the effects are individual and diluted. Switching the nudger off (08-21, post hoc) barely moved the swarm. “Messages always carry a field” is the robust, slightly deflating lesson. Prompt tweaks, room changes and most fixes showed nothing at day scale. Unconfirmed on the holdout.

## RESULTS VS PREDICTIONS

| Prediction                                                                                                                                                                                              | Round 1 $\rightarrow$ 1b (B4; V4)                                            | Verdict  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|----------|
| P1 nudges: both; $K$ 0.15 $\rightarrow$ 0.13; $\Delta\pi_{\text{idle}}$ $-0.043$ 4/5 B4; both, idle down (n.s.) $\rightarrow -0.048^*$ ; V4 both, $K$ 0.25*, 5/5 V4 $\Delta\pi_{\text{wait}}$ $-0.08^*$ |                                                                              |          |
| P2 human: field (P 0.92); chat $-0.03$ ; V4 $\times$ both, toward field/both chat                                                                                                                       |                                                                              |          |
| P3 mentions: field P 0.76 $\rightarrow$ 0.50 (neither 0.46); no mixed field to chat direction                                                                                                           |                                                                              |          |
| P4 erasure: A2 chain $K$ $-0.15$ , idle $-0.11$ , work $\times$ browse up, $+0.11$ ; V4 $K$ $+0.27$ $ K  < 0.1$                                                                                         |                                                                              |          |
| P5–P8 steps                                                                                                                                                                                             | unchanged (B4 states unaffected by as round the bin bug)                     | 1        |
| ITT (lever_design)                                                                                                                                                                                      | nudge work/chat share $+0.020$ [0.010, $-0.030$ ]                            |          |
| Native NE43                                                                                                                                                                                             | nudger off: catalytic loss (B4, V4); mixed bookends off: B4 neither, V4 both |          |
| Native NE44                                                                                                                                                                                             | 12-h escape 0.14 vs 5-min 0.40                                               | $\times$ |
| Native NE10 (V4)                                                                                                                                                                                        | neither; wait share $+0.02$                                                  | mixed    |



**Second figure.** Round 1b. (a) The lever plane: round 1 (open), round 1b on B4 (filled) and on Jev v3.1 states (triangles); nudges stay in the “both” corner in both state spaces, erasure is a field with state-space-dependent catalysis. (b) Native steps inside #51 and at NE10: percentiles of idle/wait escape and occupancy among same-shape day-boundary placebos; the nudger switch-off sits at the bottom of the escape distribution with occupancy inside the band.

## GOAL PERIODS IN DETAIL

G03 d 1b: descriptive (r1 descriptive)  
G04  $\sim$  1b: mixed (r1 supported)  
G05 d 1b: descriptive (r1 descriptive)  
G06  $\sim$  1b: mixed (r1 mixed)  
G08  $\times$  1b: failed (r1 failed)  
G10  $\times$  1b: failed (r1 failed)  
G11  $\times$  1b: failed (r1 failed)  
G12  $\times$  1b: failed (r1 failed)  
G13  $\sim$  1b: mixed (r1 mixed)  
G16  $\times$  1b: failed (r1 failed)  
+35 more periods: see the card.

## FAITHFULNESS SCORECARD

|                        |   |                   |   |
|------------------------|---|-------------------|---|
| A mapping              | 1 | F identifiability | 1 |
| B assumptions          | 1 | G ground truth    | 1 |
| C adequacy             | 1 | H comparative     | 1 |
| D unfitted predictions | 1 | I transfer        | 1 |
| E interventional       | 1 |                   |   |

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

## CAVEATS

The nudge class rests on one period (G51; G38 has the same signs but is not significant). Fields are invisible below  $\sim 100$  episodes, so “catalyst” or “neither” in small periods does not exclude a field. The field/catalyst split depends on a load-distribution convention;  $|K| < 0.10$  is not interpretable. B4 states mean different things in regimes I and III. Mentions are partly common-cause. Design choices made after seeing data are flagged: the no-consolidate erasure chain, the content-TV check, and the post hoc P9 switch-off. 28 class-level tests: the nudge field alone does not survive Bonferroni.

## ROUND 2

Run `confirm.py` after commit (expect low nudge power on #45–#50). Agent-centered content states for steps; a dose analysis of nudger re-firing; the 08-21 switch-off as a proper NE with day-matched controls.

*No round-2 redirects yet.*

H39-catalysts-vs-fields generated 2026-10-04 11:10 by in-fra/summaries/build\_summaries.py card hypotheses/H39-catalysts-vs-fields/README.md ratings: H39 round-1b agent (suggested) (page 2)